

Generative Retrieval for Cover Song Identification via Audio-Derived Semantic IDs

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Abstract—We present, to our knowledge, the first application of the generative retrieval paradigm to cover song identification (CSI). Rather than encoding a query and searching an embedding index, a T5 model autoregressively generates the Semantic ID of a cover version given the query track’s Semantic ID. We compare three constructions of audio-derived Semantic IDs: random tuples (baseline), RQ-VAE quantization of MERT-v1-95M embeddings, and the first three EnCodec RVQ codebooks used directly. Before any generative training, prefix-overlap analysis on a 3,413-track Discogs-VI-YT subset shows that MERT-RQ-VAE Semantic IDs share first-token prefixes within cliques at $4.3\times$ the random rate and $11\times$ at two-token prefixes, while EnCodec direct codes carry weaker but real structure. The structure transfers cross-dataset to Covers80 for MERT but collapses for EnCodec on low-bitrate audio, highlighting the value of a learned RQ-VAE over raw codec codes. On Discogs-VI retrieval, T5 with MERT-RQ-VAE Semantic IDs outperforms a MERT bi-encoder on five of six metrics (MRR 0.271 vs. 0.214; Recall@1 0.237 vs. 0.152), and the condition ranking matches the pre-training structure analysis exactly—confirming that audio-derived Semantic ID structure drives generative retrieval performance.

Keywords—cover song identification, generative retrieval, semantic IDs, music information retrieval, MERT, EnCodec

I. INTRODUCTION

Cover song identification (CSI) is the task of retrieving recordings of the same underlying composition under arbitrary changes of key, tempo, instrumentation, and production [8]–[10]. Standard practice frames CSI as nearest-neighbor search in a learned audio embedding space [11], [12].

Recent work in information retrieval has demonstrated that an autoregressive transformer can replace the traditional encode-and-search pipeline by generating a target document’s discrete identifier directly [1], [2]. The model’s parameters *are* the index. Adoption in music has so far targeted text-to-track and playlist continuation [3]–[5]; no published work has applied it to cover song identification.

We argue CSI is an ideal testbed for audio-derived generative retrieval. Ground truth is clean and unambiguous (two versions of “Yesterday” are positives; the rest of the corpus is not), the task is unsolved (embedding-based systems must encode compositional identity while ignoring surface acoustic

variation), and open datasets exist at scale. If audio-derived Semantic IDs (SemIDs) encode compositional structure—harmonic, melodic, tonal content—rather than surface timbre, covers of the same composition should land in nearby regions of the SemID space and share token prefixes.

The architectural observation motivating this work is that the RQ-VAE used to construct Semantic IDs in recommendation [2] and the residual vector quantizer in neural audio codecs such as EnCodec [7] are the same mechanism applied at different stages of the pipeline. This raises a natural question: can EnCodec’s codes serve directly as Semantic IDs, or is a learned RQ-VAE over a higher-level audio representation (such as MERT [6]) necessary?

Our contributions are:

- The first formulation of CSI as generative retrieval over a discrete Semantic ID vocabulary.
- A systematic comparison of three SemID constructions—random, RQ-VAE-over-MERT, EnCodec direct codes—on Discogs-VI-YT and Covers80.
- Empirical evidence that audio-derived Semantic IDs carry retrieval-relevant structure prior to any generative training, with within/cross-clique prefix-overlap ratios up to $11\times$.
- A negative finding on EnCodec direct codes for low-bitrate audio that motivates the learned-quantizer approach.

II. RELATED WORK

A. Generative Retrieval

Generative retrieval replaces the encode-and-search pipeline with a single sequence model that emits a target’s identifier directly: the model’s parameters become the index. DSI [1] casts document retrieval as text-to-identifier generation and trains with a joint indexing-and-retrieval objective. TIGER [2] introduced Semantic IDs constructed via RQ-VAE over item embeddings; its headline finding—that structured codes substantially outperform arbitrary identifiers—is the result our SemID-condition comparison tests in the audio

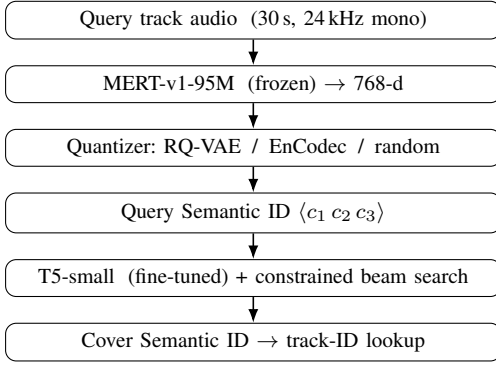


Fig. 1. Generative retrieval pipeline. A frozen audio encoder and a quantizer map each track to a discrete Semantic ID; a fine-tuned T5 autoregressively generates the Semantic ID of a cover, which is resolved to a track. The model’s parameters replace the embedding index.

domain. Music-domain extensions include Text2Tracks [3] (text prompt to track), FusID [4] (multimodal Semantic IDs for playlist continuation), and large-scale catalog work at industry labs [5]. All target recommendation or text-query settings; none addresses cover song identification, where the query and target are both audio and ground truth is unambiguous.

B. Cover Song Identification

Standard benchmarks are Covers80 [10] (80 cliques, 160 tracks), Da-TACOS [9] (25k tracks), and the recent Discogs-VI [8] (493k versions across 98k cliques, with paired YouTube IDs for audio crawl). System families include similarity-matrix alignment over CQT/HPCP features and learned bi-encoders such as MOVE [11] and CoverHunter [12]. Recent work also explores music language models (e.g. MERT [6]) as universal feature extractors.

C. Discrete Audio Representations

EnCodec [7] is a neural audio codec built around residual vector quantization (RVQ): each frame’s representation is encoded as a sequence of code indices into hierarchical codebooks. RQ-VAE [13], [14] is the same residual quantization construction, in our setting applied to the 768-dimensional MERT embedding of a 30-second clip rather than to acoustic frames. The architectural identity between these two mechanisms is exploited in our `encodec_ids` condition.

III. METHOD

A. Task Formulation

For each clique $\mathcal{C} = \{t_1, t_2, \dots, t_n\}$ in the training corpus, we generate all ordered pairs (t_i, t_j) , $i \neq j$. The model receives the Semantic ID of a query track t_q and emits the discrete Semantic ID $\text{SID}(t_j)$ of a cover (Fig. 1). Retrieval is performed by constrained beam search; predicted SIDs are resolved to track IDs via a lookup table. A prediction is correct if the resolved track lies in the query’s clique.

B. Semantic ID Construction

A Semantic ID is a 3-token tuple $(c_1, c_2, c_3) \in [0, K]^3$. We compare three constructions:

Random IDs. For each track in the corpus, sample $c_i \sim \text{Uniform}(0, K=256)$ independently. Tests the upper bound on what a generative model can memorize without structural prior.

MERT-RQ-VAE IDs. We extract a 768-d embedding per track from MERT-v1-95M [6] by mean-pooling hidden states of layers 7–12 across a 30-second center clip resampled to 24 kHz mono. We train a 3-level RQ-VAE on these embeddings via sequential mini-batch k -means. Writing the MERT embedding as the level-0 residual r_0 , level ℓ assigns the nearest codebook centroid and forwards the remaining residual:

$$c_\ell = \arg \min_k \|r_{\ell-1} - e_\ell[k]\|_2, \quad r_\ell = r_{\ell-1} - e_\ell[c_\ell], \quad (1)$$

for $\ell \in \{1, 2, 3\}$ with codebook size $K = 256$. The Semantic ID is (c_1, c_2, c_3) . Codebooks are trained on Discogs-VI and reused to encode Covers80, so the two datasets share a code space.

EnCodec direct IDs. We encode each track with EnCodec [7] at 3.0 kbps (4 codebooks of size 1024). We take the first three codebooks’ frame-level codes and aggregate to track level via majority vote, producing a 3-token SemID over $K = 1024$.

C. Generative Retrieval Model

We use T5-small (60M parameters) [15]. We extend the tokenizer with $3K$ SemID tokens—one per (level, code) pair—and resize the model’s input and output embedding matrices. The encoder input is the prefix “retrieve cover:” followed by the query’s SemID token sequence; the decoder generates the target SemID. Training uses teacher forcing with cross-entropy loss.

One implementation detail proved decisive. The standard mean-initialization for resized embeddings sets every new SemID token to the same vector; the encoder then cannot distinguish queries and the model collapses to a constant output. We instead sample each SemID embedding from the original vocabulary’s per-dimension Gaussian, which restores input sensitivity and is necessary for the model to train at all.

At inference, we apply constrained beam search of width 20 with four diverse beam groups [16]: decoder position i is restricted to the K tokens of level i , and end-of-sequence is forced after three tokens. Each beam decodes to one SemID, resolved through a precomputed lookup table to zero or more tracks; the ranked union of beam-resolved tracks (excluding the query) is the retrieved list.

D. Evaluation Regimes

Generative retrieval memorizes a fixed corpus, so the split design matters. We evaluate two regimes. *Transductive* (the standard DSI/TIGER setting): every clique is in the indexed corpus and the train/val/test split is over (query $\rightarrow$ cover) pairs. *Inductive* (cold-start): whole cliques are held out, so test

songs are entirely unseen. The contrast measures how much of generative retrieval’s performance depends on having indexed the target catalog.

IV. EXPERIMENTS

A. Datasets

Discogs-VI-YT [8]. From the 98,785-clique metadata (release dated 2024-07-01), we sampled 2,500 cliques with ≥ 2 versions (12,006 candidate YouTube IDs). Crawling with `yt-dlp` in May 2026 recovered 3,413 tracks; of the remainder, 28% were unavailable (deletions, takedowns, region locks) and 44% triggered YouTube’s anti-bot challenge. Audio was standardized to 30-second center clips at 24 kHz mono. Filtering to cliques with ≥ 2 surviving versions yields **696 usable cliques across 3,192 tracks**. The transductive split holds out (query $\rightarrow$ cover) pairs per clique (42,620 train pairs, 2,324 distinct test queries); the cold-start split holds out whole cliques (487/104/105 train/val/test).

Covers80 [10]. 80 cliques $\times$ 2 versions = 160 tracks (32 kbps mp3). *The model never sees any Covers80 track during training*; Covers80 is used exclusively as a cross-dataset held-out evaluation set.

B. Training Configuration

T5-small is fine-tuned per condition with AdamW (lr 5×10^{-4} , weight decay 0.01), batch size 64, dropout 0.3, warmup 200 steps, mixed-precision bf16, up to 15 epochs with early stopping (patience 3) on validation loss. The same hyperparameters are used for random, MERT-RQ-VAE, and EnCodec conditions; only the SemID source and codebook size differ.

C. Baselines and Metrics

The reference baseline is a MERT + FAISS **bi-encoder**: 30-second-clip MERT embeddings are L2-normalized, indexed with `IndexFlatIP`, and queried for top-20 nearest neighbors. We report MR1 (mean rank of first correct cover), MRR (mean reciprocal rank), MAP, and Recall@{1,5,10}.

V. RESULTS AND ANALYSIS

A. Semantic ID Structure Before Training

The headline empirical question is whether audio-derived SemIDs encode the right kind of similarity. We measure, for each pair of tracks, whether they share their first SemID token (p_1), their first two tokens (p_{12}), or all three (p_{123}). Rates are computed separately for within-clique pairs and a uniform sample of 100,000 cross-clique pairs. Random IDs predict $p_1 \approx 1/K$ for both, with no structure-driven gap; audio-derived IDs should produce $p_1^{\text{within}} \gg p_1^{\text{cross}}$ if covers cluster in code space.

Figure 2 and Table I summarize the result on Discogs-VI. MERT-RQ-VAE SemIDs share the first prefix **4.3 $\times$** as often within a clique as across, and the first two prefixes **11 $\times$** as often. EnCodec direct codes show a smaller but consistent signal (1.8 $\times$ and 7 $\times$). Random IDs are within sampling noise of unity. *This structure exists prior to any T5 training*—it is a property of the SemID construction itself.

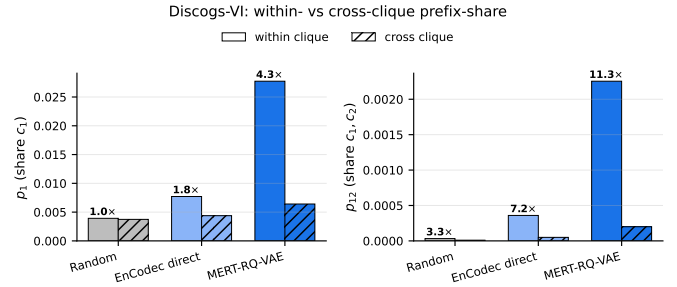


Fig. 2. Within- vs cross-clique prefix-share rates on Discogs-VI for the three SemID conditions. Bar height is the fraction of pairs sharing the indicated prefix; annotated multiplier on each within bar is the within/cross ratio. Random IDs sit at the noise floor (1.06 $\times$); MERT-RQ-VAE encodes compositional structure most strongly.

TABLE I
SEMANTIC ID QUALITY ON DISCOGS-VI (3,413 TRACKS). WITHIN/CROSS IS THE RATIO OF PREFIX-SHARE RATES INSIDE VS. ACROSS CLIQUES. HIGHER = MORE COMPOSITIONAL STRUCTURE ENCODED.

Condition	Clash	Util. c1	W/C p_1	W/C p_{12}
Random	0.0003	256/256	1.06 $\times$	3.3 $\times$
EnCodec direct	0.008	578/1024	1.8 $\times$	7.0 $\times$
MERT-RQ-VAE	0.014	256/256	4.3 $\times$	11 $\times$

B. Cross-Dataset Structure Transfer

On Covers80, encoded with the same RQ-VAE codebooks trained on Discogs-VI, MERT-RQ-VAE retains $p_1^{\text{within}}/p_1^{\text{cross}} = 1.6\times$. The signal is weaker (Covers80 has 80 within-pairs vs. 30,603 on Discogs-VI, so the variance is high), but the direction is consistent: the SemID structure is not a Discogs-VI artifact.

Figure 3 visualizes the SemID space for the 20 most-populous cliques, projected via t-SNE on the codebook-expanded vectors. Clique coloring is visible as locally coherent regions of the embedding, consistent with the prefix-overlap statistics above.

C. EnCodec Collapses on Low-Bitrate Audio

The same EnCodec direct codes that show 1.8 $\times$ structure on Discogs-VI clean WAVs collapse on Covers80’s 32 kbps mp3s: codebook utilization drops to 57/30/29 of 1024 per level, clash rate jumps to 0.41 (95 unique SemIDs for 160 tracks), and within $\approx$ cross prefix-overlap (1.0 $\times$ at p_1). The low-bitrate spectral content evidently maps to a narrow region of EnCodec’s code space, regardless of compositional identity. This is a concrete weakness of using raw codec codes as Semantic IDs and a motivation for the learned-RQ-VAE approach.

D. Main Retrieval Results

We evaluate generative retrieval in the transductive regime standard to DSI and TIGER: the full corpus is indexed and the train/val/test split is over (query $\rightarrow$ cover) pairs. Table II reports retrieval on the Discogs-VI test set (2,324 query tracks).

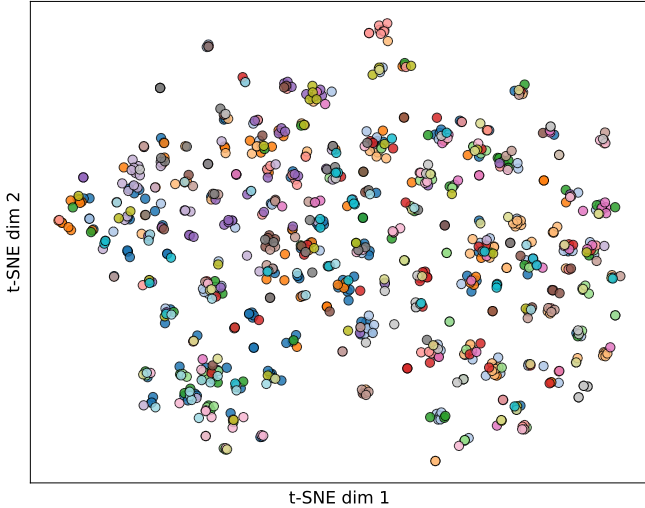


Fig. 3. t-SNE projection of MERT-RQ-VAE Semantic IDs (codebook-expanded) for the 20 most-populous Discogs-VI cliques. Tracks within a clique are colored alike; visible coloring suggests that the SemID code space separates cliques to a meaningful extent.

TABLE II

RETRIEVAL ON DISCOGS-VI (2,324 QUERY TRACKS). BI-ENCODER IS MERT + FAISS; T5 ROWS ARE GENERATIVE RETRIEVAL, ONE MODEL PER SEMID CONDITION. BEST PER COLUMN IN BOLD.

Model	MR1	MRR	MAP	R@1	R@5	R@10
Bi-encoder (MERT)	14.0	0.214	0.181	0.152	0.278	0.355
T5 random	18.1	0.112	0.100	0.095	0.122	0.148
T5 EnCodec direct	17.9	0.122	0.106	0.102	0.140	0.150
T5 MERT-RQ-VAE	13.3	0.271	0.254	0.237	0.311	0.345

Three findings. First, **generative retrieval with MERT-RQ-VAE Semantic IDs outperforms the MERT bi-encoder** on five of six metrics, losing only Recall@10 by 0.010. The Recall@1 gain is the largest—0.237 vs. 0.152, a 56% relative improvement—indicating that constrained autoregressive generation yields a sharper top-of-list ranking, while the bi-encoder catches up only in the deep tail.

Second, the condition ordering (random $\approx$ EnCodec $\ll$ MERT-RQ-VAE) is identical to the pre-training prefix-overlap ordering in Table I. The Semantic ID structure measured before training directly predicts downstream retrieval quality. T5 with random IDs reaches MRR 0.11 by pure memorization of the corpus; structured MERT IDs more than double that.

Third, this replicates the central finding of TIGER [2]—structured identifiers far outperform arbitrary ones—and demonstrates it holds for *audio-derived* structure on a real cover song retrieval task.

E. The Transductive Boundary: a Cold-Start Experiment

Generative retrieval memorizes a fixed corpus—the model’s parameters are the index. To measure this boundary directly, we re-ran all three conditions with an *inductive* clique-level split: 487/104/105 train/val/test cliques, with test cliques en-

TABLE III

COLD-START (INDUCTIVE) RETRIEVAL ON DISCOGS-VI: 105 TEST CLIQUES ENTIRELY HELD OUT FROM TRAINING. THE BI-ENCODER HAS NO TRAINING PHASE; T5 COLUMNS ARE GENERATIVE RETRIEVAL.

Model	MR1	MRR	R@1	R@10
Bi-encoder (MERT)	15.0	0.193	0.146	0.308
T5 random	20.6	0.000	0.000	0.000
T5 EnCodec direct	15.4	0.000	0.000	0.000
T5 MERT-RQ-VAE	17.7	0.000	0.000	0.000

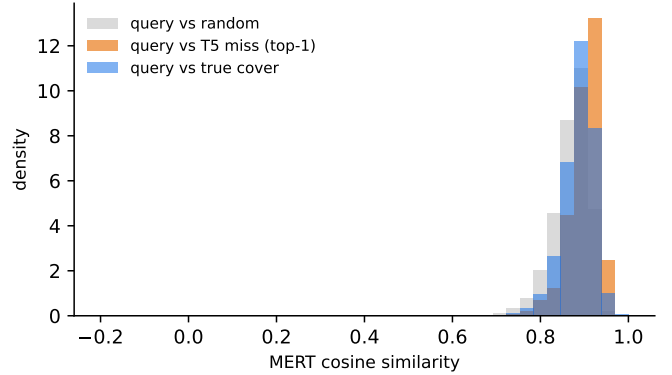


Fig. 4. Distribution of MERT cosine similarity between a query and: a random track (floor), T5’s top-1 prediction on a miss, and the query’s true covers. T5’s misses are acoustically plausible—and in fact *more* similar to the query than the true covers.

tirely unseen during training (same training recipe otherwise). Table III reports the result.

All three generative conditions score **exactly zero** on every retrieval metric. The models are not degenerate—they emit 29–59 distinct tracks across the 390 queries—but every generated Semantic ID belongs to a *training* clique. The model cannot reach a clique it never indexed. The bi-encoder, having no training phase, retrieves held-out covers exactly as well as in-distribution ones.

The contrast with Table II is the paper’s central scoping result: **generative retrieval for CSI wins when the corpus is fixed and indexed (MRR 0.271 vs. 0.214) and fails completely on new songs (0.000)**. Replacing the search index with model parameters is the source of both the transductive gain and the inductive failure. For an open or growing catalog, the bi-encoder remains necessary; for a fixed catalog, generative retrieval is the stronger choice.

F. Error Analysis: What the Model Retrieves When It Fails

For the transductive MERT-RQ-VAE model we split the 2,324 test queries into hits (867) and misses (1,457) and measured MERT cosine similarity between the query and three references (Fig. 4):

The query-to-true-cover similarity is 0.886; query-to-random is 0.869; and query-to-T5-miss is **0.900**. Two observations follow. First, the model degrades gracefully: a wrong prediction (0.900) is far closer to the query than a random track (0.869), so failures are plausible near-misses, not noise.

Second, and more revealing, T5’s misses are *more* acoustically similar to the query than the true covers are. The model fails by retrieving acoustic near-duplicates. This is direct evidence that MERT-RQ-VAE Semantic IDs encode acoustic surface similarity, which only approximates compositional identity: covers involving heavy reinterpretation—substantial key, tempo, or instrumentation change—are acoustically distant from the query and are systematically missed. Closing this gap is the key open problem for audio-derived generative retrieval.

VI. LIMITATIONS AND FUTURE WORK

Corpus scale. The Discogs-VI subset is constrained by YouTube link rot and bot detection ($\sim 72\%$ loss from a 12k-track crawl). The 3,192-track corpus is two orders of magnitude smaller than the catalogs in TIGER-style recommendation work. Recovering bot-detected entries with authenticated cookies is straightforward and would multiply the eval set; we report the un-recovered subset for reproducibility. A larger corpus would also test whether the transductive advantage scales.

The transductive constraint. Our central negative result—zero cold-start performance—is the sharpest limitation. A generative retriever cannot serve a growing catalog without retraining or an explicit indexing objective that teaches new track identities. A DSI-style indexing task [1], interleaving (audio $\rightarrow$ own-SemID) examples with the retrieval objective, is the natural next step and could partially relax the constraint.

Acoustic vs. compositional similarity. The error analysis shows the model retrieves acoustic near-duplicates when it fails. MERT-RQ-VAE Semantic IDs inherit MERT’s acoustic bias; covers with large key, tempo, or instrumentation change fall outside it. Semantic IDs built from explicitly compositional features (harmonic progression, melodic contour) could close this gap.

Modeling choices. The model trains on 30-second center clips, ignoring long-form structure, and uses only the SemID-to-SemID input mode; the continuous-MERT-to-SemID variant via a learned encoder projection is left to future work.

VII. CONCLUSION

We have introduced the first generative retrieval system for cover song identification. Audio-derived Semantic IDs from a learned RQ-VAE over MERT embeddings carry strong compositional structure ($4.3\times/11\times$ within/cross prefix-overlap), while EnCodec direct codes carry weaker and quality-sensitive structure. A 60M-parameter T5 generating MERT-RQ-VAE Semantic IDs outperforms a MERT bi-encoder on Discogs-VI retrieval (MRR 0.271 vs. 0.214), with the three SemID conditions ranking exactly as their pre-training structure predicts. The approach is transductive—it cannot retrieve un-indexed catalogs—which is the principled cost of replacing the search index with model parameters. Future work includes recovering the bot-blocked portion of the crawl for a larger corpus, the continuous-MERT-to-SemID input variant, and an indexing objective that could relax the transductive constraint.

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